Getting Started with R and RStudio

Sebastian A. J. Lortz

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University of Groningen

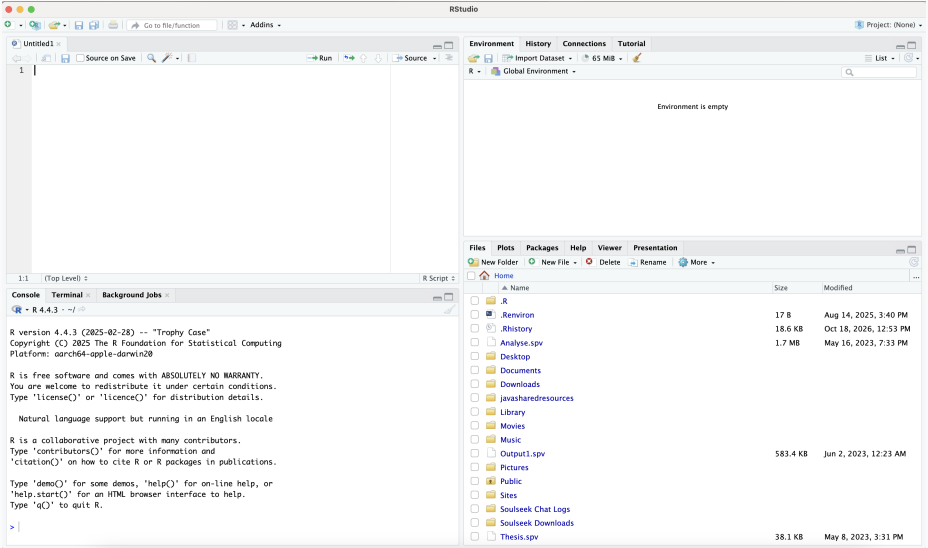
Introduction to R and RStudio

R and RStudio

- **R**: programming language and software environment for statistical computing and graphics
- Free and open source; extended with **packages** (e.g., from CRAN)
- **RStudio**: integrated development environment (**IDE**) for R, developed by Posit
- RStudio does not compute; it passes your commands to R
- Today: base R only, no packages needed

Division of labour: R computes, RStudio is the workspace around it.

The RStudio interface



Scripts, not the console

- **R script**: plain-text file (`.R`) with commands, run from top to bottom
- Run line or selection: *Ctrl* + *Enter* (Mac: *Cmd* + *Enter*)
- Run whole script: *Ctrl* + *Shift* + *Enter*
- **Comment**: text after # is not executed
- Objects live in memory and are lost on closing → the script is the permanent record

Reproducibility: same script + same data = same output.

Working directory

- **Working directory**: folder where R reads and writes files by default
- Set: *Session* > *Set Working Directory* > *Choose Directory*
- Check: `getwd()`
- File names such as `"pupils.csv"` are read relative to this folder

Error "cannot open file": almost always a wrong path name.

Core syntax

Element	Example	Meaning
Assignment	`x <- c(3, 7, 9)`	Store values in an **object**
Function call	`mean(x, na.rm = TRUE)`	**Function** applied to **arguments**
Vector	`c(3, 7, 9)`	Ordered values of one type
Data frame	`pupils$age`	Rows = cases, columns = variables; `$` selects a column
Missing value	`NA`	"Not available"
Comment	`# note`	Ignored by R

→ Help on any function: `?mean`

The example study

Evaluation of a classroom programme for social-emotional learning in primary school; 120 pupils, 60 per group.

Variable	Type	Description
`group`	Categorical	Control or intervention
`age`	Whole number	Age in years (8 to 12)
`behaviour`	Whole number	Teacher-rated behaviour problems (higher = more problems)
`wellbeing`	Whole number	Self-reported wellbeing (higher = better); some missing

Workshop roadmap

Step	Task	Main functions
1	Load the data	`read.csv()`
2	View the data	`View()`, `head()`, `str()`
3	Plot the data	`hist()`, `plot()`
4	Descriptive statistics	`summary()`, `mean()`, `sd()`
5	Group comparison plot	`tapply()`, `boxplot()`
6	t-test	`t.test()`
7	Inspect missing values	`is.na()`, `colSums()`, `table()`
8	Plot missingness on covariates	`boxplot()`

→ Save `pupils.csv` and `workshop.R` in one folder, set it as working directory, open `workshop.R` and work through the steps.

Hands-on: the workshop script

Step 1: Load the data

```
# stringsAsFactors = TRUE: text columns become categorical variables (factors)
pupils <- read.csv("pupils.csv", stringsAsFactors = TRUE)
dim(pupils)   # number of rows and columns
#> [1] 120   4
```

→ `pupils.csv` must be in the working directory; the Environment now holds `pupils`.

Step 2: View the data

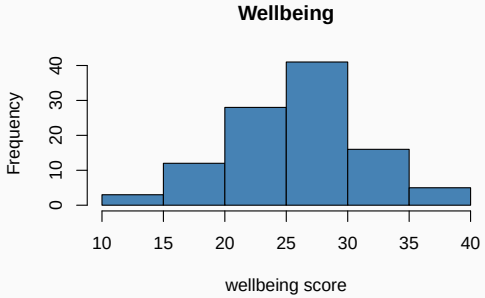
```
View(pupils)   # spreadsheet view in a new tab
```

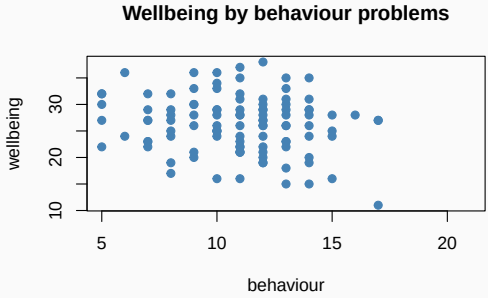
```
head(pupils)   # first six rows
#>     group age behaviour wellbeing
#> 1 control  12         7        23
#> 2 control   8         6        24
#> 3 control   8         7        22
#> 4 control  12        11        26
#> 5 control  10         9        26
#> 6 control  11        17        11
str(pupils)    # structure: type of each variable
#> 'data.frame':    120 obs. of  4 variables:
#>  $ group    : Factor w/ 2 levels "control","intervention": 1 1 1 1 1 1 1 1 1 1 ...
#>  $ age      : int  12 8 8 12 10 11 11 12 11 9 ...
#>  $ behaviour: int  7 6 7 11 9 17 12 9 10 15 ...
#>  $ wellbeing: int  23 24 22 26 26 11 24 28 25 NA ...
```

→ `Factor` = categorical, `int` = whole number, `NA` = missing value.

Step 3: Plot the data

```
hist(pupils$wellbeing, col = "steelblue",
     main = "Wellbeing", xlab = "wellbeing score")
plot(wellbeing ~ behaviour, data = pupils, pch = 19, col = "steelblue",
     main = "Wellbeing by behaviour problems")
```





→ More behaviour problems, lower wellbeing.

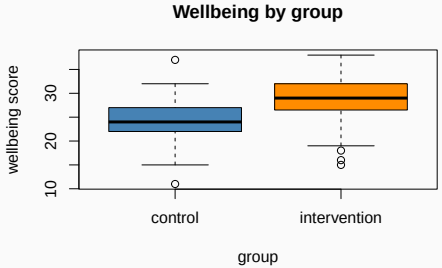
Step 4: Descriptive statistics

```
summary(pupils)
#>           group         age          behaviour       wellbeing
#>  control     :60   Min.   : 8.00   Min.   : 5.00   Min.   :11.00
#>  intervention:60   1st Qu.: 9.00   1st Qu.: 9.75   1st Qu.:23.00
#>                    Median :10.00   Median :12.00   Median :27.00
#>                    Mean   :10.15   Mean   :11.62   Mean   :26.23
#>                    3rd Qu.:12.00   3rd Qu.:14.00   3rd Qu.:29.00
#>                    Max.   :12.00   Max.   :21.00   Max.   :38.00
#>                                                    NA's   :15
mean(pupils$wellbeing)                  # NA: one missing value is enough
#> [1] NA
mean(pupils$wellbeing, na.rm = TRUE)    # na.rm = TRUE: ignore missing values
#> [1] 26.22857
sd(pupils$wellbeing, na.rm = TRUE)
#> [1] 5.394543
```

→ `summary()` reports `NA's`; most functions need `na.rm = TRUE`.

Step 5: Group comparison plot

```
tapply(pupils$wellbeing, pupils$group, mean, na.rm = TRUE)   # group means
#>      control intervention
#>     24.13208     28.36538
boxplot(wellbeing ~ group, data = pupils, col = c("steelblue", "darkorange"),
        main = "Wellbeing by group", ylab = "wellbeing score")
```



→ Intervention group higher on average; similar spread.

Step 6: t-test

```
t.test(wellbeing ~ group, data = pupils)   # Welch two-sample t-test (default)
#>
#>  Welch Two Sample t-test
#>
#> data:  wellbeing by group
#> t = -4.3486, df = 101.03, p-value = 3.277e-05
#> alternative hypothesis: true difference in means between group control and group intervention is not equal to 0
#> 95 percent confidence interval:
#>  -6.164448 -2.302170
#> sample estimates:
#>      mean in group control mean in group intervention
#>                   24.13208                   28.36538
```

- Intervention 4.2 points higher (95% CI 2.3 to 6.2), $p < .001$; true effect: $+4$
- Only complete cases used: $n = 105$ of 120

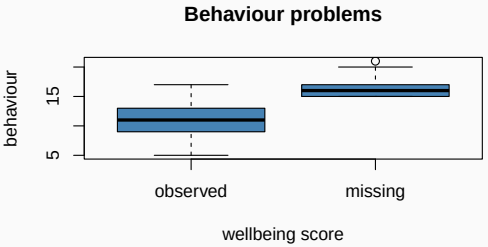
Step 7: Inspect missing values

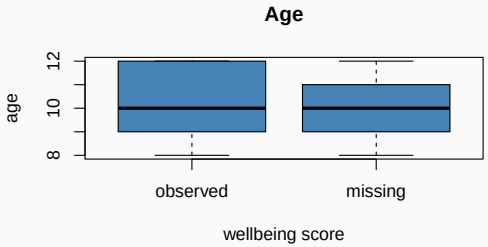
```
colSums(is.na(pupils))                          # number of NAs per variable
#>     group       age behaviour wellbeing
#>         0         0         0        15
mean(is.na(pupils$wellbeing))                   # share of NAs in wellbeing
#> [1] 0.125
table(pupils$group, is.na(pupils$wellbeing))  # TRUE = missing
#>
#>                FALSE TRUE
#>   control         53    7
#>   intervention    52    8
```

→ 15 of 120 `wellbeing` scores missing (12.5%); similar in both groups.

Step 8: Plot missingness on covariates

```
boxplot(behaviour ~ is.na(wellbeing), data = pupils, col = "steelblue",
        names = c("observed", "missing"), xlab = "wellbeing score",
        main = "Behaviour problems")
boxplot(age ~ is.na(wellbeing), data = pupils, col = "steelblue",
        names = c("observed", "missing"), xlab = "wellbeing score",
        main = "Age")
```





→ Missingness related to `behaviour`, not to `age`: not missing completely at random

Questions and Discussion

Discussion

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